



Harish Kumar S 251801090-2025-AIDS ▾

H2

Started on	Tuesday, 25 August 2026, 5:09 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 5:09 PM
Time taken	33 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  // Divide and Conquer / Two-pointer recursive function
4  void findTwoSum(int arr[], int left, int right, int x) {
5      if (left >= right) {
6          printf("No\n");
7          return;
8      }
9
10     int sum = arr[left] + arr[right];
11
12     if (sum == x) {
13         printf("%d\n%d\n", arr[left], arr[right]);
14         return;
15     }
16
17     if (sum < x) {
18         findTwoSum(arr, left + 1, right, x);
19     } else {
20         findTwoSum(arr, left, right - 1, x);
21     }
22 }
23
24 int main() {
25     int n;
26     if (scanf("%d", &n) != 1) return 0;
27
28     int arr[n];
29     for (int i = 0; i < n; i++) {
30         scanf("%d", &arr[i]);
31     }
32
33     int x;
34     scanf("%d", &x);
35
36     findTwoSum(arr, 0, n - 1, x);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			

	Input	Expected	Got	
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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